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Carrier basket transporter having carrier baskets with increased weight carrying capacity

Abstract

A carrier basket support and stabilizing system for a carrier basket having increased weight carrying capacity is provided. The system includes a plurality of cars configured to travel along a pathway, with at least some support groups including two of the cars having a carrier basket support connected thereto. The carrier basket support has a support plate connected to the first car via a pivoting connection and connected to the second car by a pivoting and sliding connection such that the cars can pivot relative to the support plate and a distance between the first and second cars can change during traversal of curves along the pathway. A support connection on the support plate is configured for connection to a mating connection on a back support of the carrier basket to allow pivoting of the carrier basket so its support surface can remain generally horizontal during movement on the pathway.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to the field of conveyor systems for use in moving products, such as delivery of orders in chain restaurants, bars, cafeterias, or the like. More particularly, the invention relates to a system for supporting and stabilizing the carrier basket as it is moved along by the conveyor between the loading and/or unloading stations.

BACKGROUND

(2) Applicant has developed and markets an overhead conveyor system which is based on a continuous track made up into a closed loop that contains an endless “train” of small wheeled cars connected together. The track is normally suspended from the ceiling or wall and can be made to snake through a building from a loading point to an unloading point and back again. Food products to be transported need to be in packages or bags, which are placed in carrier baskets carried by some of the wheeled cars. Various means may be provided to unload or release the packages or bags from the carrier basket and the food placed into a holding tray.

(3) Another food transport system for the vertical transport of packaged food items from one floor of a restaurant to another floor is described in U.S. Pat. No. 7,708,135. The food transport system includes a conveyor that transports food-item supporting carriers along a closed-loop pathway defined by a belt or chain, to which the carriers are pivotally connected. Although the transport of food items placed on supporting carriers is convenient, the belt or chain-based conveyor system has less design flexibility in terms of space utilization and winding transport pathways.

(4) US 2021/0094761 of the Applicant is also directed to a conveyor system for the transport of packaged food items in restaurants, etc., which allows a flexible design of layouts. This provides a track-based conveyor system with supporting carriers on which food items to be transported can be placed at a loading station. The food items are then transported to a customer delivery point which is remote from the loading station. Here, the packaged items are loaded onto the carriers at loading and unloading stations. In one preferred arrangement, the carriers are defined by baskets (carrier baskets) having a series of support fingers cantilevered from a single back support that is pivotally mounted to a car that can travel in a conveyor track. The carrier support fingers are aligned with spaces located between loading/unloading station support fingers or rollers. The entire content of US 2021/0094761 is incorporated herein by reference as if fully set forth.

(5) Although the known arrangements have been successful in operation, there is an issue with the weight capacity of the carrier basket when loaded since the weight of the items being transported is transferred from the carrier basket to a single one of the wheeled cars to which the carrier is pivotally connected. Improved functionality in this regard is desired.

SUMMARY

(6) The present disclosure is directed to a carrier basket support and stabilizing system for a carrier basket having increased weight carrying capacity that is transported along a conveyor having a pathway. The system includes a plurality of cars configured to travel along the pathway, with at least some support groups including two of the cars having a carrier basket support connected thereto. Each said support group of two of the cars includes a first car and a second car. The carrier basket support has a support plate connected to the first car via a pivoting connection and connected to the second car by a pivoting and sliding connection such that the first and second cars are pivotable relative to the support plate and a distance between the first car and the second car can change during traversal of curves along the pathway. A support connection is located on the support plate configured for connection to a carrier basket on a side facing away from the pathway. The carrier basket has a back support, a product support surface extending generally transversely from the back support in a direction opposite to the pathway, and a mating connection on the back

support that is connected to the support connection.

(7) With this arrangement, the carrier basket is supported by two of the cars, providing a greater weight capacity than in the prior known system where the carrier basket was supported by a single car. This arrangement also allows for a change in the distance between the first and second cars in the curves and to compensate for the changing angles of the cars relative to one another in a horizontal plane as they traverse inner and outer curves.

(8) In one embodiment, a plurality of support rollers are located on the support plate, at least one of which is configured to contact the back support at a location spaced apart from the support connection.

(9) In one embodiment, the support connection and the mating connection provide for pivoting movement between the carrier basket and the support plate, and two of the support rollers are in contact with the back support to receive cantilevered loads from articles loaded on the product support surface of the carrier basket.

(10) In one embodiment, the pivoting connection between the first car and the support plate comprises a spherical ball joint supported on the support plate and a bolt connecting the first car to the spherical ball joint.

(11) In one embodiment, the pivoting and sliding connection between the second car and the support plate comprises a spherical pillow block bearing connected to a side of the support plate facing the pathway, and a rod fastened to the second car that extends generally parallel to the pathway. The rod extends through the spherical pillow block bearing such that the second car can pivot relative to the support plate and move closer to or further from the first car by the rod sliding in the spherical pillow block bearing.

(12) In one embodiment, the first car is the lead car and the second car is the trailing car.

(13) In one embodiment, track guide rollers are located on a side of the support plate facing the pathway. These help to transfer cantilevered loads from the support plate to the track that defines the pathway.

(14) In one embodiment, the support connection comprises a plurality of connection rollers, and the mating connection comprises a housing having a part circular track that encloses greater than 180°. The connection rollers are engaged in the part circular track such that the carrier basket is pivotable by gravity so the product support surface is maintained generally horizontal during movement of the carrier basket along the pathway.

(15) In one arrangement, the pathway includes horizontal curved portions, and the first and second cars are configured to pivot relative to one another and the support plate as the support group of cars travels around the horizontal curved portions.

(16) In one arrangement, the pathway includes vertical curved portions, and the first and second cars are configured to pivot relative to one another and the support plate as the support group of cars travels around the vertical curved portions.

(17) In one embodiment, the system further includes a stabilizer bar connected to the first car and aligned generally perpendicular to a direction of the pathway, and stabilizer rollers are connected to the stabilizer bar spaced apart from the pivoting connection that are configured to contact a side of the support plate facing the pathway. This provides for additional load transfer capability between the support plate and the first car.

(18) In a preferred embodiment, the product support surface comprises a plurality of spaced apart fingers.

(19) In one embodiment, the cars include track rollers that are configured to ride on a track that defines the pathway.

(20) In one embodiment, a flexible connector extending between the first and second cars.

(21) The various features noted above can be used alone or in various combination with the support plate having the first and second cars connected thereto that provide the enhanced load carrying ability for a carrier basket by being connected to two of the cars.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Further advantages and characteristics of the invention will become apparent by the below description of embodiments making reference to the accompanying drawings, in which:
- (2) FIG. 1 is a perspective view of a prior art conveyor system of the Applicant that uses carrier baskets that are carried along the pathway defined by a track having cars that travel along the track, with each of the carrier baskets being supported by a respective single car.
- (3) FIG. 2 is a perspective view of a carrier basket support according to the present disclosure that is configured to travel along the pathway defined by a track that is supported by two of the cars that travel along the pathway to provide increased load carrying capability.
- (4) FIG. 3 is a rear perspective view showing details of the support plate for the carrier basket support shown in FIG. 2 with connections for two of the cars being illustrated.
- (5) FIG. 4 is a rear view of the carrier basket support shown in FIGS. 2 and 3 with two of the cars connected to the support plate.
- (6) FIG. 5 is a perspective view similar to FIG. 4 showing the carrier basket support with two of the cars connected to the support plate.
- (7) FIG. 6 is a front perspective view showing the carrier basket support of FIG. 5 with the support connection on the support plate for connection to a carrier basket as well as the mating connection that is configured to be connected to the back support of the carrier basket.
- (8) FIG. 7 is a view of a portion of the carrier basket transporter including a conveyor with a pathway showing the carrier basket support and stabilizing system's support plate connected to two of the cars traveling along the pathway.
- (9) FIG. 8 is a perspective view showing the carrier basket support traversing an inside horizontal curve along the pathway.
- (10) FIG. 9 is a perspective view, in cross-section, similar to FIG. 8 showing in detail the pivoting connection between the first car and the support plate and the pivoting and sliding connection between the second car and the support plate, as well as the engagement of the cars in the track that forms the pathway via track rollers connected to the cars.
- (11) FIG. 10 is a perspective view, partially in cross-section, similar to FIG. 9, showing the carrier basket support that is connected to two of the cars traversing an outside horizontal curve along the pathway.
- (12) FIG. 11 is a top view of the arrangement shown in FIG. 9.
- (13) FIG. 12 is a perspective view showing the carrier basket support connected to a carrier basket detailing the side of the carrier basket support that faces the pathway.
- (14) FIG. 13 is a perspective view showing the carrier basket support along with a carrier basket supported thereon traversing an outside horizontal curve along the pathway.
- (15) FIG. 14 is perspective view similar to FIG. 13 showing the carrier basket support along with a carrier basket located thereon traversing an inside horizontal curve along the pathway.
- (16) FIG. 15 is a top view of the arrangement shown in FIG. 13.
- (17) FIG. 16 is a top view of the arrangement shown in FIG. 14.
- (18) FIG. 17 is a perspective view showing the carrier basket support along with a carrier basket supported thereon traversing a vertical curve along the pathway defined by a conveyor of the carrier basket transporter.
- (19) FIG. 18 is an elevational view of the arrangement shown in FIG. 17.
- (20) FIG. 19 is an enlarged perspective view showing the support connection on the support plate along with a housing of the mating connection that is configured to be connected to the back support of the carrier basket which allows the carrier basket to pivot by gravity so that the product support surface is maintained generally horizontal during movement of the carrier basket along the

pathway.

DETAILED DESCRIPTION

(21) Certain terminology is used in the following description for convenience only and is not limiting. The words “right,” “left,” “top,” and “bottom” designate directions in the drawings to which reference is made. The words “a” and “one,” as used in the claims and in the corresponding portions of the specification, are defined as including one or more of the referenced item unless specifically stated otherwise. This terminology includes the words above specifically mentioned, derivatives thereof, and words of similar import. The phrase “at least one” followed by a list of two or more items, such as “A, B, or C,” means any individual one of A, B or C as well as any combination thereof. The terms approximately or generally mean within $\pm 10\%$ of a specified value unless otherwise noted, and within $\pm 25^\circ$ of a specified angle or direction.

(22) FIG. 1 is a view showing an embodiment of Applicant's conveyor system **10** from U.S. 2021/0094761, which is incorporated herein by reference as if fully set forth. This includes a pathway **12** defined by a track **13** having one or more curved portions **14** as well as straight segments **15** in which a plurality of cars **20** are guided. A motor drives the cars **20** along the track **13** either via having the cars **20** directly linked to one another or having intermediate portions located between the cars **20**. A carrier basket **30** is pivotally mounted to at least some of the cars, with the carrier basket **30** including carrier support fingers **32** that are cantilevered from a side of the carrier basket **30** that is pivotally mounted to the cars **20**. At least one of a loading station **40** or an unloading station **42** is preferably located along the pathway **12**. The loading and unloading stations **40**, **42** include support fingers **44** that are used to remove an item being transported from the carrier basket **30** by the carrier basket support fingers **32** moving between the loading or unloading station support fingers **44** to either pick-up or drop-off the item being transported via the conveyor system **10**. These support fingers **44** can be powered, i.e., formed by powered rollers, belts or chains, in order to further carry the item along a transport or delivery path. As discussed in U.S. 2021/0094761, the conveyor system **10** allows enhanced flexibility for the design of layouts in order to suit particular applications and space requirements.

(23) Referring to FIGS. 2-19, an embodiment of a carrier basket support and stabilizing system **50** for a carrier basket **110** to provide increased weight carrying capacity is shown. The carrier basket support and stabilizing system **50** is for use in connection with a carrier basket transporter **52** that includes a conveyor **54** (see FIGS. 7-11 and 13-18), similar to the conveyor system **10** disclosed in FIG. 1, that has a pathway **56** along which the carrier baskets **110** are carried. The pathway **56** is defined by tracks **57**, similar to the track **13** discussed above, that can be arranged in various configurations having curved portions **58**, **59** as well as straight segments in order to move products, which can be for example fast food, that is placed in the carrier basket **110**, preferably between loading and unloading stations, such as **40** and **42** shown in FIG. 1.

(24) In accordance with the present system **50**, as shown in FIGS. 2-7 as well as 9-12, a plurality of cars **60** are configured to travel along the pathway **56** (see FIG. 7). At least some support groups **62** for each carrier basket **110** include two of the cars **60a**, **60b** that have a carrier basket support **80** connected thereto. Each support group **62** of two of the cars **60** includes a first car **60a** and a second car **60b**. These are shown most clearly in FIGS. 4, 5, 7, and 9-11. The cars **60**, **60a**, **60b** preferably include track rollers **68** that are configured to ride on the track **57** that defines the pathway **56**.

(25) In order to support a heavier weight for the carrier basket **110**, the carrier basket support **80** includes a support plate **82** that is connected to the first car **60a** via a pivoting connection **86** and connected to the second car **60b** by a pivoting and sliding connection **94** such that both cars **60a**, **60b** can pivot relative to the support plate **82** and a distance between a first car **60a** and a second car **60b** can change during traversal of curves along the pathway **56**, for example as shown in FIGS. 7 and 9-17.

(26) As shown in detail in FIGS. 3 and 9-11, the pivoting connection **86** between the first car **60a** and the support plate **82** preferably comprises a spherical ball joint **87** that is supported on the

support plate **82**, for example, it can be screwed or bolted thereon, as well as a bolt **88** (a bolt can refer to a threaded fastener or a pin) that connects the first car **60a** to the spherical ball joint **87**. As can be seen in particular in FIGS. **9** and **10**, the spherical ball joint **87** has an opening defined therethrough and the bolt **88** extends through the opening, preferably inside a bushing **89** that has a collar or shoulder **90** that spaces the spherical ball of the spherical ball joint **87** from a connecting surface of the first car **60a** in order to allow nonbinding pivoting movement. In the exemplary embodiment shown, a nut is engaged with the bolt in order to clamp the bushing **89** and spherical ball of the spherical ball joint **87** against a mounting surface on the first car **60a** by tightening the nut onto the end of the bolt **88**. The bolt **88** can be accessed through a clearance opening in the support plate **82**. While this type of pivoting connection is one exemplary embodiment, other types of pivoting connections could be used.

(27) Still with reference to FIGS. **3** and **9-11**, the pivoting and sliding connection **94** between the second car **60b** and the support plate **82** comprises a spherical pillow block bearing **95** which can be, for example, attached via screws or bolts to the support plate **82**, on a side of the support plate **82** faces the pathway **56**. A rod **96** that is fastened to the second car **60b**, preferably via a bracket **97**, extends generally parallel to the pathway **56**, and the rod **96** extends through the spherical pillow block bearing **95** such that the second car **60b** can pivot relative to the support plate **82** via movement of the spherical ball of the spherical pillow block bearing **95**, and can also move closer to or further from the first car **60a** by the rod **96** sliding in the opening that extends through the spherical ball of the spherical pillow block bearing **95**.

(28) As shown in detail in FIGS. **4**, **5**, and **9-11**, a flexible connector **70** extends between the first car **60a** and the second car **60b**. This can be formed of a polymeric material, and may have a form with undulations in order to allow flexing and compression or expansion as a distance between the first and second car **60a**, **60b** changes as the group **62** with the two of the cars **60a**, **60b** and the carrier basket support **80** traverse horizontal and/or vertical curves **58**, **59** along the pathway **56**.

(29) In order to provide further support, preferably a plurality of support rollers **83**, preferably at least 3 and more preferably four, are located on the support plate **82**, at least one of which is configured to contact the back support **112** at a location spaced apart from the support connection **110**. More preferably, two of the support rollers **83** are in contact with the back support **112** to support cantilevered loads from articles **111**, shown schematically in FIG. **16**, that are loaded on the product support surface **114** of the carrier basket **110**. In the illustrated embodiment, the support rollers **83** are located at the corners of the support plate **82**. This ensures that at least two of the support rollers **83** contact the back support **112** regardless of the pivoted position of the carrier basket **110** as it moves along the pathway **56**.

(30) As shown in detail in FIGS. **2-6**, **8** and **12-16**, track rollers **84** are preferably located on a side of the support plate **82** facing the pathway **56**. These track guide rollers **84** are spaced apart from the main portion of the track **57** and are adapted to contact a bottom rail and/or a top rail located along the track in order to maintain the support plate **82** generally vertically aligned as it moves along the pathway **56** and receive loads from the carrier basket **110** which can be transferred into the support plate **82** via the support rollers **83** on the support plate **82** contacting the back support **112** of carrier basket **110**.

(31) Referring now to FIGS. **2**, **6**, and **12-18**, a support connection **100** is located on the support plate **82** and is configured to connection to the carrier basket **110** on a side facing away from the pathway **56**. The carrier basket **110** includes a back support **112** as well as a product support surface **114** extending generally normal to the back support **112** in a direction opposite to the pathway **56**. A mating connection **120** is located on or otherwise connected to the back support **112** and is connected to the support connection **100** on the support plate **82**. The support connection **100** and the mating connection **120** provide for pivoting movement between the carrier basket **110** and the support plate **82**.

(32) Referring to FIGS. **2**, **5**, **6**, and **19**, in one embodiment the support connection **100** includes a

plurality of connection rollers **102**, and the mating connection **120** comprises a housing **122** having a part circular track **124** that enclosing greater than 180°, and more preferably encloses approximately 270°. The connection rollers **102** are engaged in the part circular track **124** such that the carrier basket **110** is pivotable by gravity by the connection rollers **102** allowing the part circular track **124** to roll thereon so that the product support surface **114** is maintained generally horizontal during movement of the carrier basket **110** along the pathway **56**. The connection rollers **102** which are fixed relative to the support plate **82** providing for low resistance support of the part circular track **124** in the housing **122** thereon. The housing **122** is connected to or formed as part of the back support **112**. Preferably there are a minimum of three of the connection rollers **102**, and more preferably there are four of the connection rollers **102** arranged to define a circular support that generally corresponds to the circular track **124**. The circular track **124** preferably includes a lip **126** that extends radially inwardly so that it is held in position on the connection rollers **102**. In a preferred embodiment, the housing **122** is formed separate from and includes a grooved connection **128** that is engageable with a corresponding structure on the back support **112** of the carrier basket **110**.

(33) Other types of pivoting connections could also be provided between the support plate **82** and the back support **112** of the carrier basket **110**, such as a single pin pivot support with a pin extending from one of the parts being received in an opening in the other of the parts. The opening could be defined as a simple slot, or could be in the form of a bearing to reduce friction.

(34) As shown in FIGS. **8-11** and **13-16**, the pathway **56** can include horizontal curved portions **58**, and the first and second cars **60a**, **60b** are configured to pivot relative to one another and the support plate **82** as the support group **62** with the cars **60a**, **60b** travels around the horizontal curve portions **50a**. Further, as shown in FIGS. **17** and **18**, the pathway **56** also can include vertical curved portions **59**, and the first and second cars **60a**, **60b** are configured to pivot relative to one another and the support plate **82** as the support group **62** with the cars **60a**, **60b** travels around the vertical curved portions **59**.

(35) Referring again to FIGS. **4** and **5**, in order to provide enhanced stability, a stabilizer bar **64** can optionally be connected to the first car **60a** and aligned generally normal to a direction of the pathway **56**. Stabilizer rollers **66** are connected to the stabilizer bar **64**, preferably at the ends thereof, that are spaced apart from the pivoting connection **86**. These stabilizer rollers are configured to contact a side of the support plate **82** facing the pathway **56** in order to provide additional support and alignment for the support plate **82** as the first car **60a** pivots relative to the support plate **82** while traversing the vertical curved portions **59** of the pathway **56**.

(36) In a preferred embodiment, the cars **60** include track rollers **68** that are configured to ride on the track **57** that defines the pathway **56**. However, rollers are not required and slider blocks or other low-friction type guides could be utilized.

(37) In one preferred embodiment, the first car **60a** is the lead car and the second car **60b** is the trailing car in a direction of travel of the group **60** of the cars **60a**, **60b** along the pathway. However, this could be reversed.

(38) Using one or more of the above features, higher loads can be carried by the carrier basket **110** as it is supported on two of the cars **60a**, **60b**. In a preferred embodiment, the carrier basket **110** includes the support surface **114** formed with a plurality of spaced apart fingers **116** such that it could be used in a system similar to the system **10** in FIG. **1** as disclosed Applicant's prior U.S. Patent Application Publication 2021/0094761.

(39) It will be appreciated that the foregoing is presented by way of illustration only and not by way of any limitation. It is contemplated that various alternatives and modifications may be made to the described embodiments without departing from the spirit and scope of the invention. Having thus described the present invention in detail, it is to be appreciated and will be apparent to those skilled in the art that many physical changes, only a few of which are exemplified in the detailed description of the invention, could be made without altering the inventive concepts and principles

embodied therein. It is also to be appreciated that numerous embodiments incorporating only part of the preferred embodiment are possible which do not alter, with respect to those parts, the inventive concepts and principles embodied therein. The present embodiment and optional configurations are therefore to be considered in all respects as exemplary and/or illustrative and not restrictive, the scope of the invention being indicated by the appended claims rather than by the foregoing description, and all alternate embodiments and changes to this embodiment which come within the meaning and range of equivalency of said claims are therefore to be embraced therein.

Claims

1. A carrier basket support and stabilizing system for a carrier basket transporter including a conveyor having a pathway, the system comprising: a plurality of cars configured to travel along the pathway, at least some support groups including two of the cars having a carrier basket support connected thereto, each said support group of two of the cars including a first car and a second car; the carrier basket support including: a support plate connected to the first car via a pivoting connection and connected to the second car by a pivoting and sliding connection such that the first and second cars are pivotable relative to the support plate and a distance between the first car and the second car can change during traversal of curves along the pathway; and a support connection located on the support plate configured for connection to a carrier basket on a side facing away from the pathway; the carrier basket including: a back support; a product support surface extending generally transversely from the back support in a direction opposite to the pathway; and a mating connection on the back support that is connected to the support connection; and the support connection and the mating connection form a pivot connection that provides for pivoting movement between the carrier basket and the support plate.
2. The system of claim 1, wherein the pivoting connection between the first car and the support plate comprises a spherical ball joint supported on the support plate and a bolt connecting the first car to the spherical ball joint.
3. The system of claim 1, wherein the pivoting and sliding connection between the second car and the support plate comprises a spherical pillow block bearing connected to a side of the support plate facing the pathway, and a rod that is fastened to the second car that extends generally parallel to the pathway, the rod extending through the spherical pillow block bearing such that the second car can pivot relative to the support plate and move closer to or further from the first car by the rod sliding in the spherical pillow block bearing.
4. The system of claim 1, wherein the first car is the lead car and the second car is the trailing car.
5. The system of claim 1, further comprising track guide rollers located on a side of the support plate facing the pathway.
6. The system of claim 1, wherein the support connection comprises a plurality of connection rollers, and the mating connection comprises a housing having a part circular track that encloses greater than 180°, and the connection rollers are engaged in the part circular track such that the carrier basket is pivotable by gravity so the product support surface is maintained generally horizontal during movement of the carrier basket along the pathway.
7. The system of claim 1, wherein the pathway includes horizontal curved portions, and the first and second cars are configured to pivot relative to one another and the support plate as the support group of cars travels around the horizontal curved portions.
8. The system of claim 1, wherein the pathway includes vertical curved portions, and the first and second cars are configured to pivot relative to one another and the support plate as the support group of cars travels around the vertical curved portions.
9. The system of claim 1, further comprising a stabilizer bar connected to the first car and aligned generally normal to a direction of the pathway, and stabilizer rollers connected to the stabilizer bar spaced apart from the pivoting connection that are configured to contact a side of the support plate

facing the pathway.

10. The system of claim 1, wherein the product support surface comprises a plurality of spaced apart fingers.

11. The system of claim 1, wherein the cars include track rollers that are configured to ride on a track that defines the pathway.

12. The system of claim 1, further comprising a flexible connector extending between the first and second cars.

13. A carrier basket support and stabilizing system for a carrier basket transporter including a conveyor having a pathway, the system comprising: a plurality of cars configured to travel along the pathway, at least some support groups including two of the cars having a carrier basket support connected thereto, each said support group of two of the cars including a first car and a second car; the carrier basket support including: a support plate connected to the first car via a pivoting connection and connected to the second car by a pivoting and sliding connection such that the first and second cars are pivotable relative to the support plate and a distance between the first car and the second car can change during traversal of curves along the pathway; and a support connection located on the support plate configured for connection to a carrier basket on a side facing away from the pathway; the carrier basket including: a back support; a product support surface extending generally transversely from the back support in a direction opposite to the pathway; and a mating connection on the back support that is connected to the support connection; and a plurality of support rollers located on the support plate, at least one of which is configured to contact the back support at a location spaced apart from the support connection.

14. The system of claim 13, wherein the support connection and the mating connection provide for pivoting movement between the carrier basket and the support plate, and two of the support rollers are in contact with the back support to receive cantilevered loads from articles loaded on the product support surface of the carrier basket.

15. The system of claim 13, wherein the pivoting connection between the first car and the support plate comprises a spherical ball joint supported on the support plate and a bolt connecting the first car to the spherical ball joint.

16. The system of claim 13, wherein the pivoting and sliding connection between the second car and the support plate comprises a spherical pillow block bearing connected to a side of the support plate facing the pathway, and a rod that is fastened to the second car that extends generally parallel to the pathway, the rod extending through the spherical pillow block bearing such that the second car can pivot relative to the support plate and move closer to or further from the first car by the rod sliding in the spherical pillow block bearing.

17. The system of claim 13, wherein the first car is the lead car and the second car is the trailing car.

18. The system of claim 13, further comprising track guide rollers located on a side of the support plate facing the pathway.

19. The system of claim 13, wherein the support connection comprises a plurality of connection rollers, and the mating connection comprises a housing having a part circular track that encloses greater than 180°, and the connection rollers are engaged in the part circular track such that the carrier basket is pivotable by gravity so the product support surface is maintained generally horizontal during movement of the carrier basket along the pathway.

20. The system of claim 13, wherein further comprising a stabilizer bar connected to the first car and aligned generally normal to a direction of the pathway, and stabilizer rollers connected to the stabilizer bar spaced apart from the pivoting connection that are configured to contact a side of the support plate facing the pathway.
